

FIRST INTRODUCED during NASA's Apollo missions in the 1970s, the concept of digital twins has come a long way in the past five decades. But what truly put it in the spotlight in recent times was the pandemic. Take the case of contract manufacturer Flex. When a leading global medical equipment company shortlisted Flex to manufacture its Class II diabetes product in the first quarter of 2020, it had a challenge on its hands. The US-headquartered company had to accelerate and optimise the development of the product in the wake of the Covid-19 pandemic while ensuring quality and reliability as any mistakes in the design or manufacturing process could put patients at risk. To optimise the manufacturing of this product, it created a digital twin of the factory floor. With the virtual representation of the complex production system, Flex tested how it would perform under various scenarios and gained insights into optimisation tactics. "Without simulation, this testing process would have taken at least three months longer. But our DES [discrete event simulation] tool allowed us to complete our optimisation analysis in only three weeks, enabling our customer to get the product to market faster. This saved our customer time and money," says Paul Baldassari, Executive Vice President of Worldwide Operations at Flex. Plus, there were other efficiency gains—a 70 per cent increase in station output by optimising work order sizes, 47 per cent more units produced per quarter, and a 15-21 per cent higher value-added process utilisation.

Back home, Mumbai-headquartered Tata Consultancy Services (TCS) collaborated with non-profit Prayas Help Group to develop a digital twin of Pune city to predict the evolution of the pandemic. "We were able to simulate the likely extent of virus spread as people go about their lives, and the consequent impact on health infrastructure, institutional quaran-

THE COST OF DIGITAL TWINS



Typical proofs of concept (PoC) or one-plant pilots cost \$200,000-500,000, depending on the size and scale of operations and the complexity. Plus, there is the cost of maintaining it along with the necessary IT infrastructure



Payback period is typically 6-12 months for PoCs, with 10x RoI

SOURCE: WIPRO

25

PER CENT

THE IMPROVEMENT IN EMPLOYEE PRODUCTIVITY PIRAMAL GLASS SAW AFTER ADOPTING DIGITAL TWINS

ting infrastructure, testing infrastructure and so on," says K. Ananth Krishnan, Chief Technology Officer at TCS. This work is being repurposed to help define strategies to effect work-from-home to work-from-office transition, he adds.

But what is a digital twin? In the simplest of terms, it can be understood as a computer program that uses real-world data to create simulations that can predict how a product or process will perform. It can be a real-time virtual replica of a real-world entity of say, an aircraft engine, a factory shop floor or even a human being. "A digital twin is sometimes referred to as a computer-based copy of anything that exists physically," says Praveen Mysore, Director of Industry Process Consultants and Solutions at Dassault Systèmes India. But it isn't a standalone technology and involves multiple solutions that include the Industrial Internet of Things (IIoT), connectivity, cloud computing, AI and ML.

With increasing demand from healthcare and pharma industries because of the pandemic, the global digital twin market is projected to grow at a CAGR of 58 per cent to reach \$48.2 billion by 2026 from \$3.1 billion in 2020, according to MarketsandMarkets, a research firm. While India-specific numbers aren't available, Mysore believes that because of the government's push for digitisation and smart city projects, the Asia Pacific region is likely to capture a considerable share, with China, India and Japan leading the way.

Rising popularity

Digital twins have come a long way since inception, and the evolution of technologies such as CAD, IoT, cloud, and big data analytics has made it easier and a lot cheaper to adopt. The initial version was a simple simulation-based event where organisations created the virtual versions of the physical structures to test them for various load cases. The primary advantage was the model's ability to identify the underlying risks without